



On the simultaneous inference of susceptibility distributions and intervention effects from epidemic curves

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ABSTRACT

Susceptible–Exposed–Infectious–Recovered (SEIR) models with inter-individual variation in susceptibility or exposure to infection were proposed early in the COVID-19 pandemic as a potential element of the mathematical/statistical toolset available to policy development. In comparison with other models employed at the time, those designed to fully estimate the effects of such heterogeneity tended to predict small epidemic waves and hence require less containment to achieve the same outcomes. However, these models never made it to mainstream COVID-19 policy making due to lack of prior validation of their inference capabilities. Here we report the results of the first systematic investigation of this matter in idealized scenarios created with synthetic data. We simulate datasets using the model with strategically chosen parameter values, and then conduct maximum likelihood estimation to assess how well we can retrieve the assumed parameter values. Parameter uncertainties were found to markedly reduce when concurrently fitting multiple epidemics with shared parameters, suggesting a general methodological approach that can be further developed to tackle real-world questions.

1. Introduction

Susceptible–Exposed–Infectious–Recovered (SEIR) models are used extensively to study epidemics and guide public health policies. These models range in detail, from simple systems of ordinary differential equations (ODEs), to higher-dimensional implementations that include disease progression, age structure, and other forms of heterogeneity (Diekmann et al., 2013). Metapopulation (Keeling and Rohani, 2002) or agent-based models (Kerr et al., 2021) are most commonly used when explicit descriptions of spatial connectivity are desired, although they also enable the representation of other heterogeneities such as person-to-person variability as stylized in Ayabina et al. (2025).

During the COVID-19 pandemic, several authors highlighted the significant role of inter-individual variation (heterogeneity) in susceptibility and exposure to infection in flattening epidemic curves (Britton et al., 2020; Neipel et al., 2020; Rose et al., 2021; Tkachenko et al., 2021; Montalbán et al., 2022; Gomes et al., 2022). However, despite being understood for decades (McKendrick, 1940; Gart, 1968, 1972; Ball, 1985; Katriel, 2012), these ideas were treated with caution due to alleged parameter identifiability issues (Wood et al., 2026; Gomes et al., 2026). Here we report the results of a thorough investigation of parameter identifiability in SEIR models.

We use SEIR models with heterogeneous susceptibility to infection as proposed in Montalbán et al. (2022) and Gomes et al. (2022) during the pandemic. These models include non-pharmaceutical interventions (NPIs) as required, which also tend to flatten epidemic curves and be potentially confounded with heterogeneity in susceptibility. To test our ability to attribute flattening of epidemic curves correctly to heterogeneity or NPIs we generate synthetic epidemic datasets by simulating stochastic versions of those models. Then we perform sets of statistical inferences by fitting homogeneous and heterogeneous versions of the models to the generated data. Model parameters are estimated by maximum likelihood.

2. Mathematical models

2.1. SEIR models with heterogeneous susceptibility

We adopt an SEIR model which Montalbán et al. (2022) have previously analysed and Gomes et al. (2022) have previously applied to the COVID-19 pandemic. Heterogeneity in susceptibility to infection

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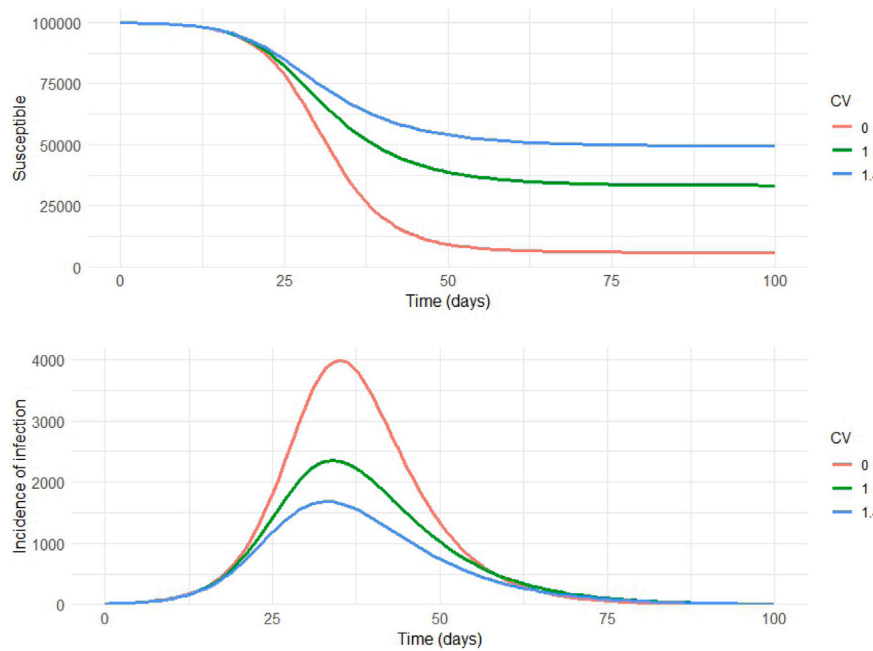


Fig. 1. Dynamics of infection under the heterogeneous susceptibility SEIR model for different values of the coefficient of variation ν . The top figure shows how numbers of susceptible individuals decrease over time, while the bottom depicts incidences of infection (epidemic curves). The homogeneous SEIR model corresponds to $\nu = 0$.

is incorporated as a multiplicative factor, x , applied to the rate of infection. The model is written in terms of differential equations as

$$\begin{aligned} \frac{dS(x)}{dt} &= -c(t) \beta (\rho E + I) \frac{x S(x)}{N}, \\ \frac{dE}{dt} &= c(t) \beta (\rho E + I) \int \frac{x S(x)}{N} dx - \delta E, \\ \frac{dI}{dt} &= \delta E - \gamma I, \end{aligned} \quad (1)$$

where $S(x)$ represents the density of susceptible individuals as a function of the susceptibility factor x , $E + I$ is the number of individuals who have been exposed and are infected (accounting for an early stage of lower infectiousness, E). The recovered number R is derived from the conservation of the total population size, $S + E + I + R = N$, where $S = \int S(x) dx$. The main parameters are the average effective contact rate β , the rate of progression from E to I (assumed $\delta = 1/5.5$ per day [McAloon et al., 2020](#); [Lauer et al., 2020](#)), the removal rate from I (assumed $\gamma = 1/4$ per day [Nishiura et al., 2020](#); [Li et al., 2020](#)) and the reduced infectiousness while in E (assumed $\rho = 0.5$). An additional time-dependent parameter $c(t)$ is included, which has a default value of 1 and will be used to study the effect of interventions. Initial conditions for variables $S(x, t)$, $E(t)$, $I(t)$ are defined as to satisfy $S(x, 0) = (1 - 3.5 \epsilon) q(x) N$, $E(0) = 2.5 \epsilon N$ and $I(0) = \epsilon N$ (see Supplementary Material Section S1 for justification), where $q(x)$ is a probability density function with mean 1 and coefficient of variation

$$\nu = \sqrt{\int (x - 1)^2 q(x) dx}. \quad (2)$$

The basic reproduction number is the dominant eigenvalue of the next generation operator ([Diekmann et al., 1990](#)), which, conditionally on the population having mean susceptibility 1, is given by

$$\mathcal{R}_0 = \beta \left(\frac{\rho}{\delta} + \frac{1}{\gamma} \right). \quad (3)$$

Following [Novozhilov \(2008\)](#) and [Montalbán et al. \(2022\)](#) showed that when $q(x)$ is a gamma distribution, system Eq. (1) can be reduced

exactly to

$$\begin{aligned} \frac{dS}{dt} &= -c(t) \beta (\rho E + I) \left(\frac{S}{N} \right)^{1+\nu^2}, \\ \frac{dE}{dt} &= c(t) \beta (\rho E + I) \left(\frac{S}{N} \right)^{1+\nu^2} - \delta E, \\ \frac{dI}{dt} &= \delta E - \gamma I. \end{aligned} \quad (4)$$

The derivation relies on a property of the gamma distribution, whereby as individuals are removed through infection, the mean susceptibility over those that remain susceptible reduces according to the formula

$$\frac{\bar{S}(t)}{S(t)} = \left(\frac{S(t)}{N} \right)^{\nu^2}, \quad (5)$$

but the distribution remains gamma. Furthermore, the gamma distribution was described by [Rose et al. \(2021\)](#) as a “natural choice to account for variations in susceptibility” as it is the limiting distribution to which other initial distributions of susceptibility converge. This choice will be kept throughout this paper, although the treatment of the explicit system Eq. (1) can be replicated for any distribution while the use of the reduced version Eq. (4) is specific to gamma distributed susceptibility. In either formulation, the classical SEIR model is retrieved by setting $\nu = 0$.

Previous studies, including those in [Novozhilov \(2008\)](#), [Neipel et al. \(2020\)](#), [Gomes et al. \(2022\)](#), [Montalbán et al. \(2022\)](#) and [Rose et al. \(2021\)](#), have demonstrated that heterogeneity in susceptibility reduces epidemic sizes. This occurs because highly susceptible individuals are infected first and hence selected out of the susceptible pool (a dynamic which is not captured by the classical homogeneous model). This effect is illustrated in [Fig. 1](#) (top). For example, around 50% of the population is expected to be infected over the course of an outbreak when CV is $\nu = \sqrt{2} \approx 1.414$. By comparison, when a homogeneous model is assumed ($\nu = 0$) the expected infected percentage rises to around 90%. In the bottom panel, corresponding trajectories for incidence of infection (represented as δE) are shown, highlighting the higher peak in the absence of heterogeneity.

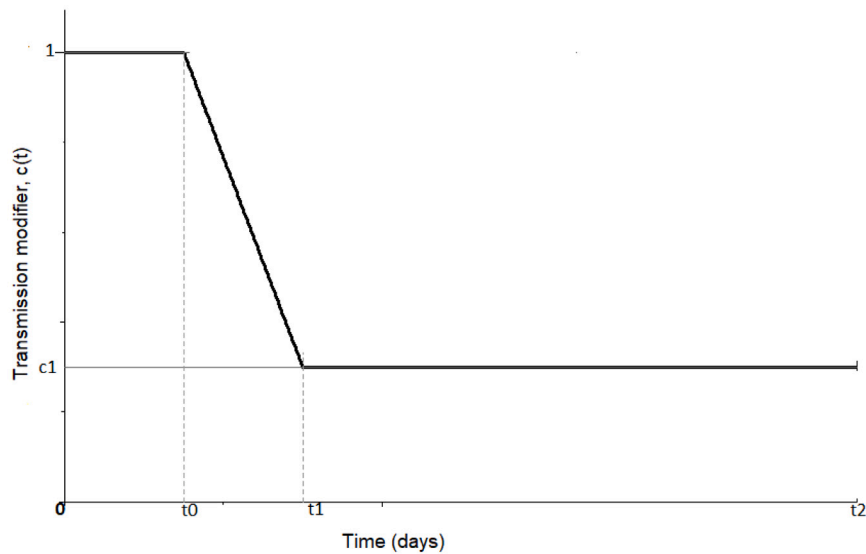


Fig. 2. Schematic illustration of the factor $c(t)$, representing the combined effects of adaptive behavioural changes and NPIs on transmission.

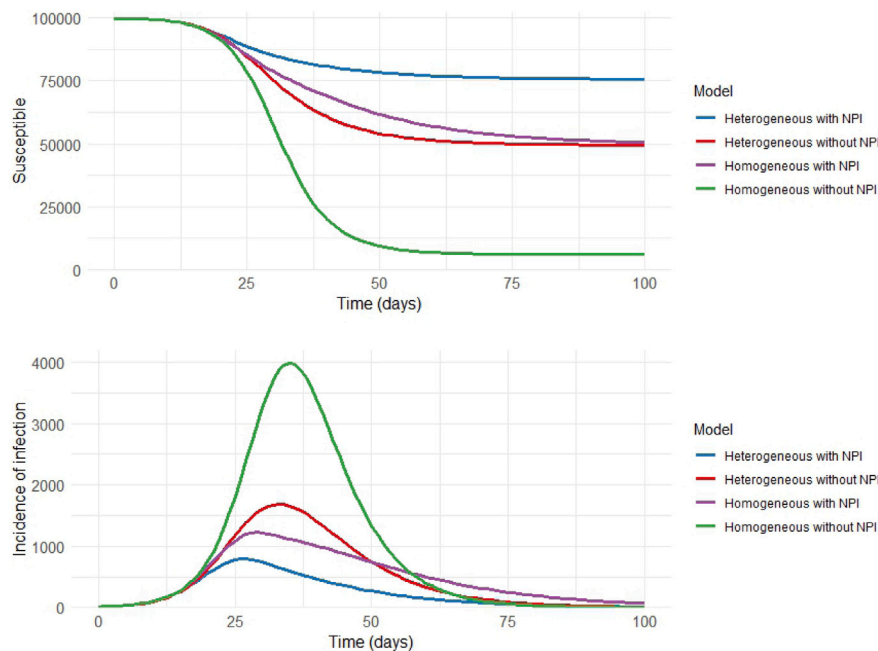


Fig. 3. The combined effects of heterogeneity in susceptibility and a hypothetical NPI with $c_1 = 0.4$ on susceptible numbers (top) and epidemic trajectories (bottom) for different values of ν . The homogeneous SEIR model corresponds to $\nu = 0$.

2.2. Non-pharmaceutical interventions (NPIs)

Outbreak awareness triggers, to some extent, a reduction in contact rates. This may be due to voluntary precautions or restrictions imposed by governments. To incorporate the effects of these so-called non-pharmaceutical interventions (NPIs) we use a time dependent factor $c(t)$ such as

$$c(t) = \begin{cases} 1, & \text{if } 0 < t \leq t_0 \\ 1 - (1 - c_1) \frac{t - t_0}{t_1 - t_0}, & \text{if } t_0 < t \leq t_1 \\ c_1, & \text{if } t_1 < t \end{cases} \quad (6)$$

where $0 \leq c_1 \leq 1$. This is profiled in Fig. 2, with t_0 representing the time when contact rates begin to decrease and t_1 marking the beginning of maximal containment (such as lockdown).

When $c_1 < 1$, the dynamics of the susceptible numbers and the incidence of infection for a homogeneous model are affected as shown in Fig. 3. It is clear that both heterogeneity in susceptibility and NPIs flatten epidemic curves although for the specific NPI profiles adopted here the flattened curves have distinctive shapes.

A description of all the model parameters is provided in Table 1.

3. Simulations

We aim to identify the effects of heterogeneity in susceptibility and NPIs on epidemic trajectories based on a given epidemic dataset affected by both. We design a systematic study with four different incidence time series generated as follows:

- D1. Homogeneous model ($\nu = 0$) without NPIs ($c_1 = 1$).
- D2. Heterogeneous model ($\nu > 0$) without NPIs ($c_1 = 1$).

Table 1

Description of model parameters. Notice that $\mathcal{R}_0$ is linearly related to the average effective contact rate β via Eq. (3), and hence only one of the two is needed to specify the model. We chose $\mathcal{R}_0$ given its more interpretable meaning.

Parameter	Description	Value
N	Total population	100,000
δ	Rate of progression from E to I	1/5.5 per day
γ	Rate of removal from I	1/4 per day
ρ	Infectiousness in E (relative to I)	0.5
$\mathcal{R}_0$	Basic reproduction number	Variable
ν	Coefficient of variation in susceptibility	Variable
c_1	Maximal reduction in transmission by NPIs	Variable
t_0	Time to beginning of behavioural change	15 days
t_1	Time to beginning of lockdown	20 days

D3. Homogeneous model ($\nu = 0$) with NPIs ($c_1 < 1$).

D4. Heterogeneous model ($\nu > 0$) with NPIs ($c_1 < 1$).

The datasets are generated by running the deterministic model Eq. (4), with NPIs as in Eq. (6), and assuming a Poisson distribution for the observations. We define incidence of infection as δE and adopt a Poisson error structure to obtain the time series

$$y(t) \sim \text{Poisson}(\hat{y}(t)) \quad \text{for each day } t \in \mathbb{N}.$$

For each combination of parameters, 200 datasets are simulated and we fit both homogeneous and heterogeneous models to estimate the respective sets of parameters as follows:

- M1. Homogeneous model ($\nu = 0$) without NPIs ($c_1 = 1$). Parameters fitted ($\mathcal{R}_0$).
- M2. Heterogeneous model ($\nu > 0$) without NPIs ($c_1 = 1$). Parameters fitted ($\mathcal{R}_0, \nu$).
- M3. Homogeneous model ($\nu = 0$) with NPIs ($c_1 < 1$). Parameters fitted ($\mathcal{R}_0, t_0$ and c_1).
- M4. Heterogeneous model ($\nu > 0$) with NPIs ($c_1 < 1$). Parameters fitted ($\mathcal{R}_0, \nu, t_0$ and c_1).

Other parameters are fixed as in Table 1. All fittings are done over 100 days.

4. Maximum likelihood estimation

With the heterogeneous susceptibility models, we estimate two sets of parameters from the datasets described in the previous section:

$$\theta_a := (\mathcal{R}_0, \nu) \quad (7)$$

$$\theta_b := (\mathcal{R}_0, \nu, t_0, c_1). \quad (8)$$

The first case corresponds to estimating θ_a by fitting models with no NPIs (M1, M2) to datasets D1, D2, as defined in Section 3. This is to establish how well heterogeneity can be estimated in the absence of the potentially confounding effects of interventions, and to test how reliable the heterogeneous susceptibility model is at determining that there is no heterogeneity when the dataset has been generated by the homogeneous model.

The second case of interest is estimating θ_b by fitting the NPI models (M3, M4) to datasets D3, D4. In addition, we shall also fit M3, M4 to D1, D2 to determine whether the inference correctly identifies the absence of interventions.

Let $x_1, x_2, \dots, x_n$ be the model simulation of the observations for the time series epidemic data $y_1, y_2, \dots, y_n$. We assume that the observations y_i are conditionally independent given the model states x_i . The likelihood function is then defined as the product of the individual probability densities:

$$\mathcal{L}(\theta) := \prod_{i=1}^n f(y_i | x_i, \theta). \quad (9)$$

where $f(\cdot)$ is the probability mass function (Poisson or negative binomial) and θ is the vector of estimated parameters.

It is usually convenient to work with the log-likelihood function given by

$$l(\theta) := \log \mathcal{L}(\theta) = \sum_{i=1}^n f(y_i | x_i, \theta). \quad (10)$$

Parameter estimation was performed in Rstudio by maximizing the log-likelihood function Eq. (10), representing the probability of observing the simulated data $\{y_i\}$ given the model realizations $\{x_i\}$.

5. Baseline analysis

This section presents a baseline simulation-inference study for a representative choice of parameters. We consider four simulated scenarios (D1–D4) and fit four models (M1–M4), as described in Section 3. The aims are to assess whether the model is correctly selected by evaluating quality of fits by the Akaike information criterion (AIC), and how parameter estimates behave.

Table 2 reports MLE summaries and 95% Wald confidence intervals. In scenarios with $c_1 = 1$ (no intervention), t_0 is redundant and its Wald interval can become numerically unstable (marked †). When c_1 is close to 1, Wald intervals on the natural scale can also be poorly behaved and should be interpreted with caution (†).

5.1. Homogeneous datasets without NPIs (D1)

Illustrative fittings for this case are provided in Supplementary Figure S2.

We first fit the no-NPI models to assess whether heterogeneity is correctly inferred as absent when the data are truly homogeneous. Fitting M1 (homogeneous, no NPIs) and M2 (heterogeneous, no NPIs) to D1 yields essentially identical estimates of $\mathcal{R}_0$ and very similar AIC values (rows (D1,M1) and (D1,M2) in Table 2). The heterogeneous model returns $\hat{\nu}$ close to 0 and the AIC difference between M1 and M2 is small, indicating no strong evidence that the added heterogeneity parameter improves fit for truly homogeneous data.

We next fit the NPI models M3 and M4 to the same D1 datasets (rows (D1,M3) and (D1,M4)). In this over-parameterized setting (since the true data have $c_1 = 1$), the models place $\hat{c}_1$ close to 1, as expected, but $\hat{t}_0$ becomes redundant and its Wald interval becomes meaningless (†). Accordingly, the large reported upper bounds for t_0 in these rows should not be interpreted as evidence for a late intervention; rather, they reflect a near-flat likelihood when the fitted model allows an intervention that is not present in the data. The AIC values remain close across M3 and M4 in this scenario.

5.2. Homogeneous datasets with NPIs (D3)

This scenario is especially informative because both heterogeneity and NPIs can flatten epidemic curves. Illustrative fittings are shown in Supplementary Figure S3.

Fitting the NPI models M3 (homogeneous with NPIs) and M4 (heterogeneous with NPIs) to D3 datasets yields very similar estimates for $\mathcal{R}_0, t_0$, and c_1 (rows (D3,M3) and (D3,M4) in Table 2). However, although the true data are homogeneous ($\nu = 0$), the heterogeneous model M4 can attribute part of the intervention effect to heterogeneity, returning a nonzero $\hat{\nu}$ with a wide confidence interval. This bias is visible in Fig. 4. The AIC values for (D3,M3) and (D3,M4) are similar, indicating that goodness-of-fit alone may not reliably distinguish whether flattening is due to NPIs or heterogeneity in this single-epidemic setting.

These results foreshadow the confounding challenge addressed in later sections: when both mechanisms are available to the fitted model, single-epidemic data can permit partial trade-offs between ν and NPI parameters (t_0, c_1) without a decisive change in AIC.

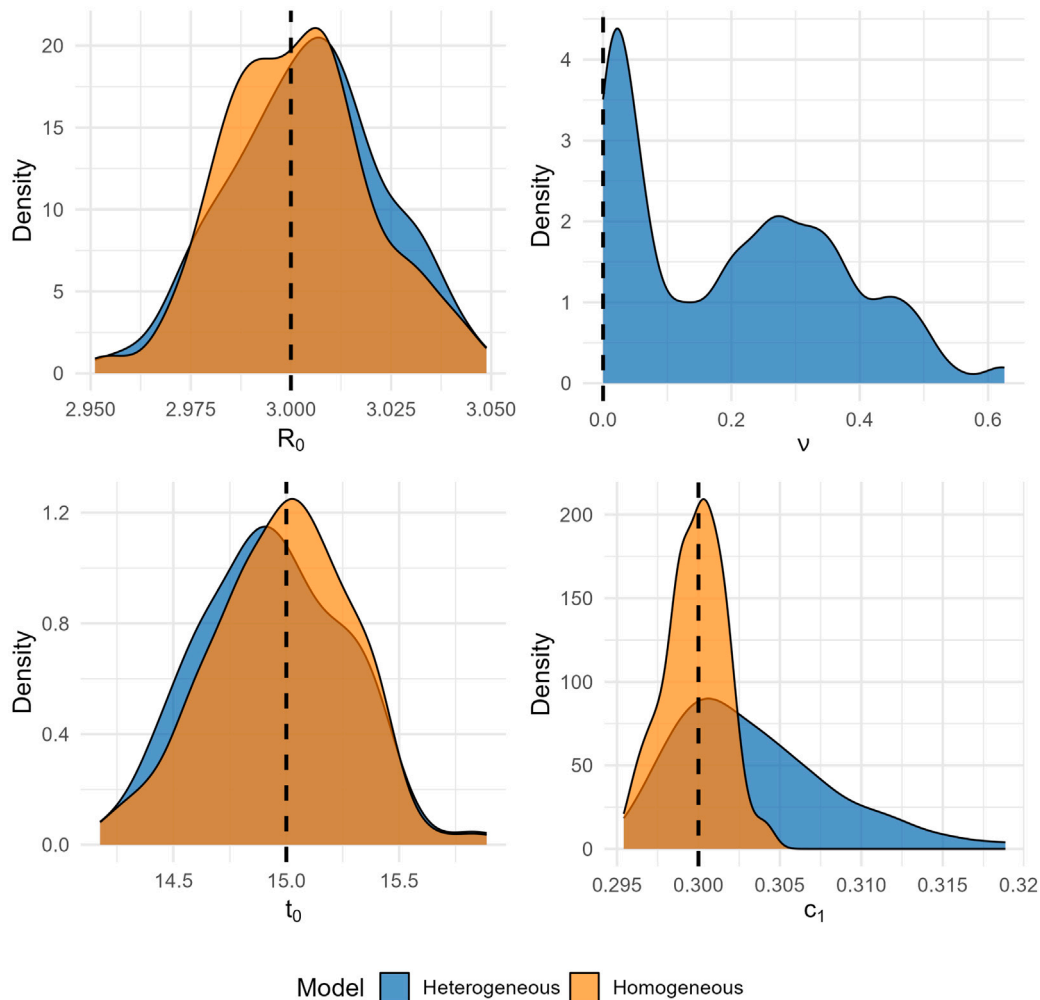


Fig. 4. Distributions for the estimated parameters for D3 datasets: Homogeneous data with impact of NPIs ($R_0 = 3$, $c_1 = 0.3$, $t_0 = 15$ days). The shaded distributions represent estimates from the heterogeneous susceptibility model, while the orange represent those from the homogeneous model. Vertical dashed lines indicate the true specified parameter values. Both models accurately recover R_0 , t_0 , and the value of the NPI parameter c_1 , and the heterogeneous model estimates a coefficient of variation ν near 0. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Strong correlations also appear between R_0 , t_0 (D4,M3), in inverse-Hessian correlation structures (Fig. 5). In Supplementary Material Section S3 we investigate how these correlations vary with the initial condition $I(0)$ and intervention strength parameter c_1 .

Understanding the uncertainty in parameter estimates is also crucial for predicting the potential course of outbreaks. Using the estimated parameters for both homogeneous and heterogeneous models, we constructed prediction intervals and examined the models' forecasting capabilities. We used a multivariate normal distribution for generating parameter sets that preserve the correlations between parameter estimates. This approach allows us to assess not only how well models fit observed data but, crucially, how reliably they predict future epidemic trajectories.

Fig. 6 shows predictions of how epidemic trajectories might respond to the complete removal of NPIs. We performed extended simulations with NPIs lifted from day 100 onwards, so that a second wave of infections could occur. Once containment is removed, $c(t)$ returns to 1, and, in all scenarios considered here, sufficient individuals have remained susceptible to fuel a new peak. In addition to accurately capturing the trajectory of the epidemic in the fitting period (days 1–100, before and during NPIs), both models are also good at forecasting beyond that (days 100–250). This is despite the heterogeneous model having wrongly estimated a small but non-zero coefficient of variation. The

wide confidence band for the heterogeneous model is mainly due to the uncertainty in the estimation of CV, which has a 95% CI of (0, 0.4, 0.912).

5.3. Heterogeneous datasets without NPIs (D2)

Illustrative fittings for this scenario are provided in Supplementary Figure S4.

We first fit the no-NPI models to assess whether heterogeneity is recoverable when interventions are absent. Comparing rows (D2,M1) and (D2,M2) in Table 2, the heterogeneous no-NPI model M2 accurately recovers both R_0 and ν and achieves a much lower AIC than the homogeneous no-NPI model M1. The homogeneous model attempts to compensate for the missing heterogeneity by shifting R_0 , but the resulting fit remains poor, as reflected by the very large AIC for (D2,M1). This indicates that, when NPIs are not permitted in the fitted model, heterogeneity produces a trajectory shape that cannot be matched well by a homogeneous model through R_0 adjustment alone.

We next fit the NPI models (M3 and M4) to the same D2 datasets (rows (D2,M3) and (D2,M4)). Model M4 (heterogeneous with NPIs) still recovers R_0 and ν well and estimates c_1 close to 1, indicating no intervention, while the inferred t_0 is redundant ($\dagger$). In contrast, the homogeneous NPI model M3 can achieve an improved fit relative to M1 by introducing a false intervention (estimating $c_1 < 1$ and t_0), thereby

Table 2

Baseline simulation-inference combinations. Each row corresponds to a scenario of simulated Dx datasets and fitted model My. Entries are representative MLEs with 95% Wald confidence intervals in brackets.

Dataset	Fitted	Parameter estimates (95% CI)				
		R_0	ν	t_0	c_1	AIC
D1	M1	3.00 (3.00, 3.01)	NA	NA	NA	815.87
	M2	3.00 (2.99, 3.01)	0.067(0.017, 0.249)	NA	NA	816.90
	M3	3.00 (2.99, 3.00)	NA	0.00(1.3 e-3, 1.335) ^a	0.999 (0.959, 1.00) ^b	818.85
	M4	3.00 (2.99, 3.03)	0.08(0.217, 0.245)	1.11 (0.687, 1.685) ^a	0.997 (0.98, 1.00) ^b	816.47
D2	M1	2.90 (2.89, 2.91)	NA	NA	NA	30237.71
	M2	3.00 (2.99, 3.01)	1.415 (1.40, 1.43)	NA	NA	811.76
	M3	2.74 (2.73, 2.75)	NA	28.48 (28.26, 28.69)	0.352 (0.348, 0.355)	2137.56
	M4	3.00 (2.98, 3.05)	1.411 (1.389, 1.423)	1.75 (1.02, 2.30) ^a	0.983 (0.893, 1.00) ^b	824.10
D3	M3	3.00 (2.97, 3.04)	NA	15.0 (14.4, 15.60)	0.30 (0.296, 0.304)	791.09
	M4	3.00 (2.97, 3.04)	0.201 (0.04, 0.912)	14.92 (14.32, 15.56)	0.302 (0.296, 0.312)	793.66
D4	M3	2.94 (2.89, 3.00)	NA	14.91 (14.3, 15.5)	0.24 (0.236, 0.243)	728.57
	M4	3.00 (2.95, 3.05)	1.406 (1.157, 1.768)	15.00 (14.37, 15.65)	0.299 (0.274, 0.331)	712.87

Simulated datasets: D1: homogeneous ($\nu = 0$), no NPIs ($c_1 = 1$); D2: heterogeneous ($\nu > 0$), no NPIs ($c_1 = 1$); D3: homogeneous ($\nu = 0$), with NPIs ($c_1 < 1$); D4: heterogeneous ($\nu > 0$), with NPIs ($c_1 < 1$).

Fitted models: M1: homogeneous, no NPIs (fit R_0); M2: heterogeneous, no NPIs (fit R_0, ν); M3: homogeneous, with NPIs (fit R_0, t_0, c_1); M4: heterogeneous, with NPIs (fit R_0, ν, t_0, c_1).

^a When the simulated data have no intervention ($c_1 = 1$), the change-point time t_0 is redundant because the transmission modifier does not change over time. In these cases the log-likelihood surface is effectively flat in the t_0 direction, so inverse-Hessian (Wald) confidence intervals for t_0 can become redundant or unstable.

^b When c_1 is at or near the boundary ($c_1 \approx 1$), symmetric Wald intervals on the natural scale are known to be poorly calibrated. We therefore interpret these fits primarily through whether the optimizer places c_1 close to 1 (no intervention) and summarize uncertainty calibration across replicated datasets in Section 5.5.

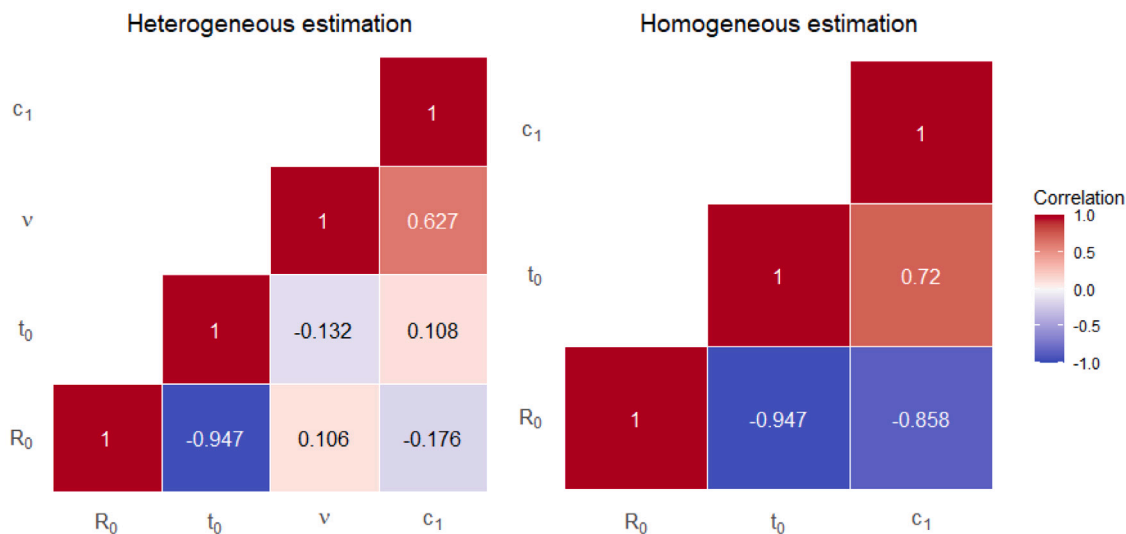


Fig. 5. Median parameter correlation heatmaps for D3 datasets: homogeneous data with impact of NPIs. Correlations between pairs of estimated parameters are provided in scenarios where fittings are performed with a heterogeneous model (M4, left) or with a homogeneous model (M3, right). In both cases, it is noteworthy the very strong negative correlation between R_0 and t_0 (-0.947).

misattributing heterogeneity-driven flattening to NPIs. Nevertheless, the AIC remains substantially worse for (D2,M3) than for (D2,M4), indicating that allowing heterogeneity is strongly favoured when the true data are heterogeneous even if the fitted model also permits NPIs.

5.4. Heterogeneous datasets with NPIs (D4)

This is the most elaborate dataset, combining both heterogeneous susceptibility and interventions. Illustrative fittings are shown in Supplementary Figure S5.

Comparing rows (D4,M3) and (D4,M4) in Table 2, the heterogeneous NPI model M4 recovers all parameters with estimates close to the simulation values, and achieves the lower AIC. The homogeneous NPI model M3 underestimates R_0 and compensates by estimating a stronger intervention (lower c_1), illustrating the same trade-off mechanism seen in D2 under misspecification.

Strong correlations also appear between R_0 and t_0 (D4,M3), and between ν and c_1 (D4,M4), in inverse-Hessian correlation structures (Fig. 8). In Supplementary Material Section S3 we investigate how these correlations vary with the initial condition $I(0)$ and intervention strength parameter c_1 , while in Section 6 we show how they can be substantially alleviated by concurrent epidemic inferences which enable higher precision of estimates. In Supplementary Material Section S4 we explore the sensitivity of the inferences reported here to population size. To test the validity of using the deterministic approximation in our parameter identifiability analysis, we have replicated the most elaborate scenario (D4, M4) using a stochastic process model (tau-leaping with a fixed time step as in Bjørnstad (2018)). The conclusions from the inference remain essentially unchanged (Supplementary Material Section S5).

Fig. 9 concerns predictions of how epidemic trajectories might respond to the complete removal of NPIs. While both models fit the

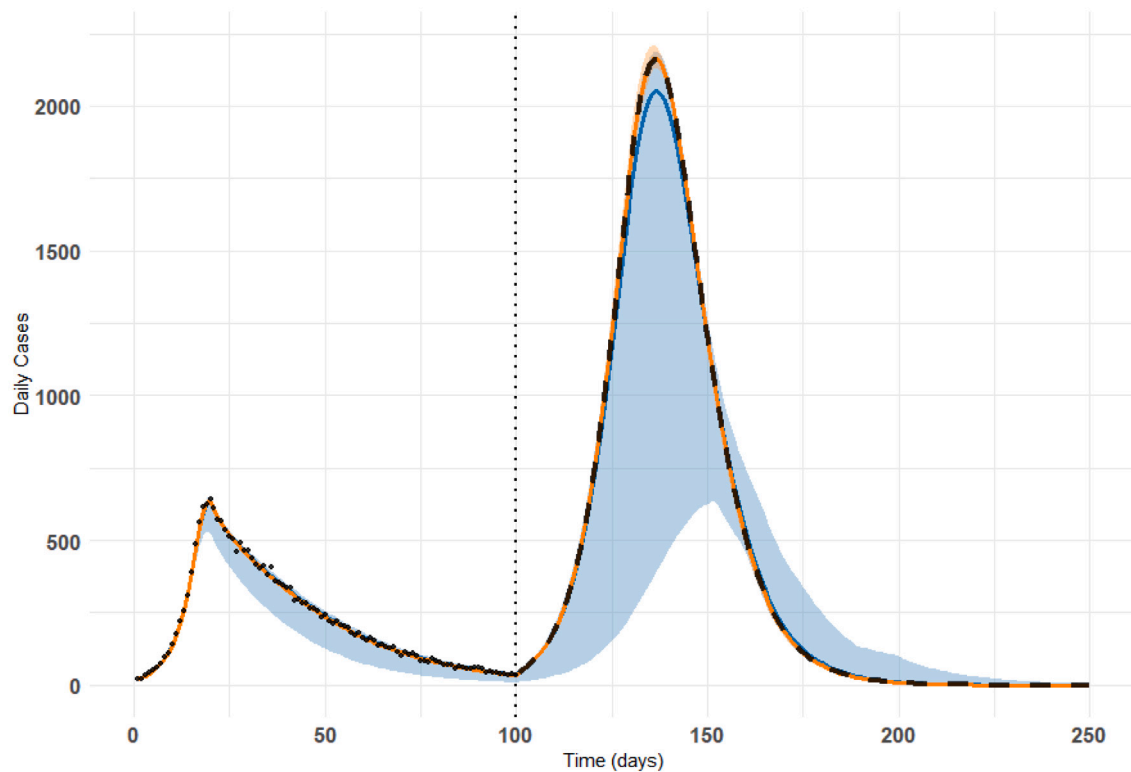


Fig. 6. Prediction trajectories for scenario D3: homogeneous data with NPIs. Models were fitted to the first 100 days of data (black dots) and then used to forecast the next 150 days following the lifting of NPIs at day 101. Heterogeneous model fit (M4, blue); homogeneous model fit (M3, orange). Lines are median trajectories and shaded areas represent 95% CI. The dashed black line represents the “true” trajectory generated with the assumed parameter values. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

initial 100 days of data reasonably well (before and during NPIs), their forecasts diverge substantially once contacts return to normal. The homogeneous model predicts a massive second wave with peak daily cases approaching 2750 — nearly four times higher than first wave. In contrast, the heterogeneous model predicts a more modest second wave with peak daily cases between 300 and 1000, approximately.

When compared to the “ground truth” (dashed black line), it is evident that the heterogeneous model’s forecast is remarkably accurate, while the homogeneous model’s prediction represents a substantial overestimation. This difference in forecasting performance stems from the homogeneous model’s inability to account for the depletion of highly susceptible individuals during the first wave, which naturally limits the severity of subsequent waves. Instead, the homogeneous model attributes the flattening of the curve in the first wave primarily to intervention effects, incorrectly suggesting greater susceptibility of the population for the second wave.

The wider prediction intervals for the heterogeneous model reflect uncertainty in the estimation of CV but, despite this uncertainty, the upper bound of its prediction interval remains well below the homogeneous model’s median forecast. This underscores that accounting for population heterogeneity is essential not only for accurate parameter estimation but also for reliable forecasting of epidemic trajectories.

5.5. Uncertainty quantification across replicated datasets (D3 and D4)

Two notions of uncertainty are used in our simulation study. First, the density plots (Figs. 4, 7) summarized the empirical sampling distribution of the maximum-likelihood estimates (MLEs) across 200 independently simulated datasets, capturing between-dataset variability for the cases (D3,M3) and (D3,M4), and (D4,M3) and (D4,M4), respectively. Second, for each dataset we compute within-dataset Wald uncertainty using the inverse observed information matrix, obtained

by inverting the Hessian of the negative log-likelihood evaluated at the MLE (in a transformed parameterization to respect parameter constraints). Following this procedure, we report two diagnostics aggregated across the replicated datasets: (i) the relative bias of the median MLE across replicates; and (ii) a typical within-dataset 95% Wald interval width (mean UCL–LCL across replicates). Because we are mainly concerned with confounding between NPIs and heterogeneity, we report these diagnostics only for the two baseline scenarios simulated with NPIs (D3 and D4) in Table 3.

In cases where the true value of a parameter is zero (notably ν in D3), relative bias is undefined; in such cases we report the absolute bias instead.

6. Fitting pairs of epidemics to overcome identifiability issues

As highlighted in Section 5, we have encountered correlations between estimated parameters. Given our focus on heterogeneity in susceptibility, correlations that involve the coefficient of variation are of special interest (particularly those between ν with the non-pharmaceutical intervention parameter, c_1).

To address this problem, we draw on a simple yet effective idea. If given a single epidemic, characterized by a set of parameter values and initial conditions, we have difficulties inferring the relative effects of CV and NPIs, we want to ask what might the trajectory look like had the same NPIs started at an earlier or later time point (so that we can observe the effect of CV alone in one realization while CV+NPIs are both effective in the other). Although we cannot conduct this experiment beyond simulation studies, it is often not difficult in real scenarios to find suitable epidemic trajectories which can be paired with our focus epidemic to alleviate identifiability issues.

This principle was employed by Gomes et al. (2022), who fitted epidemic models jointly to COVID-19 data from England and Scotland.

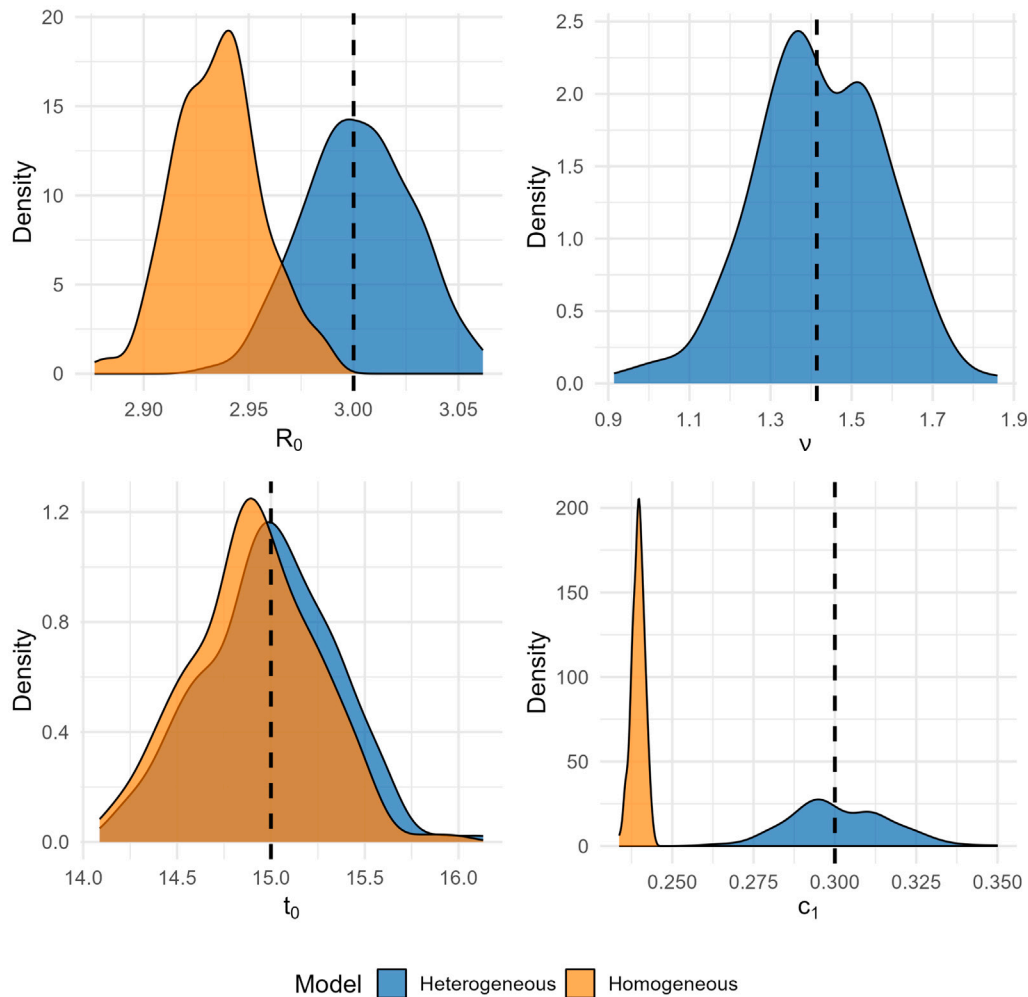


Fig. 7. Distributions for the estimated parameters for D4 datasets: Heterogeneous data with impact of NPIs ($R_0 = 3$, $v = 1.414$, $c_1 = 0.3$, $t_0 = 15$ days). The shaded distributions represent estimates from the heterogeneous susceptibility model, while the orange represent those from the homogeneous model. Vertical dashed lines indicates the true specified parameter values. The heterogeneous model accurately recovers all parameters, while the homogeneous model underestimates R_0 the NPI parameter (predicting $c_1 \approx 0.24$ instead of the true value 0.3).

Table 3

Uncertainty and bias summaries across 200 replicated datasets for the two NPI baseline scenarios: D3 (homogeneous truth with NPIs) and D4 (heterogeneous truth with NPIs). For each fitted model, SD is the standard deviation of the MLE across datasets (between-dataset variability). CI width is the mean 95% Wald interval width (UCL–LCL) computed within each dataset and averaged across datasets (within-dataset uncertainty). Bias is computed from the median MLE across datasets; relative bias is reported as a percentage where defined, otherwise absolute bias is reported.

Dataset	Parameter	M3 — homogeneous fit			M4 — heterogeneous fit		
		Bias	SD	CI width	Bias	SD	CI width
D3 homogeneous	R_0	0.491%	0.018	0.069	0.492%	0.018	0.073
	v	—	—	—	0.263 ^a	0.154	0.622
	t_0	1.629%	0.307	1.141	1.685%	0.304	1.215
	c_1	0.485%	0.002	0.007	1.767%	0.005	0.017
D4 heterogeneous	R_0	2.145%	0.021	0.073	0.687%	0.025	0.097
	v	—	—	—	9.071%	0.160	0.605
	t_0	1.847%	0.336	1.166	1.860%	0.352	1.287
	c_1	20.160%	0.002	0.008	4.051%	0.015	0.057

^a For D3, the true $v = 0$, so relative bias is undefined. We therefore report the absolute bias $|\text{median}(\hat{v}) - 0| = 0.263$ instead of a percentage.

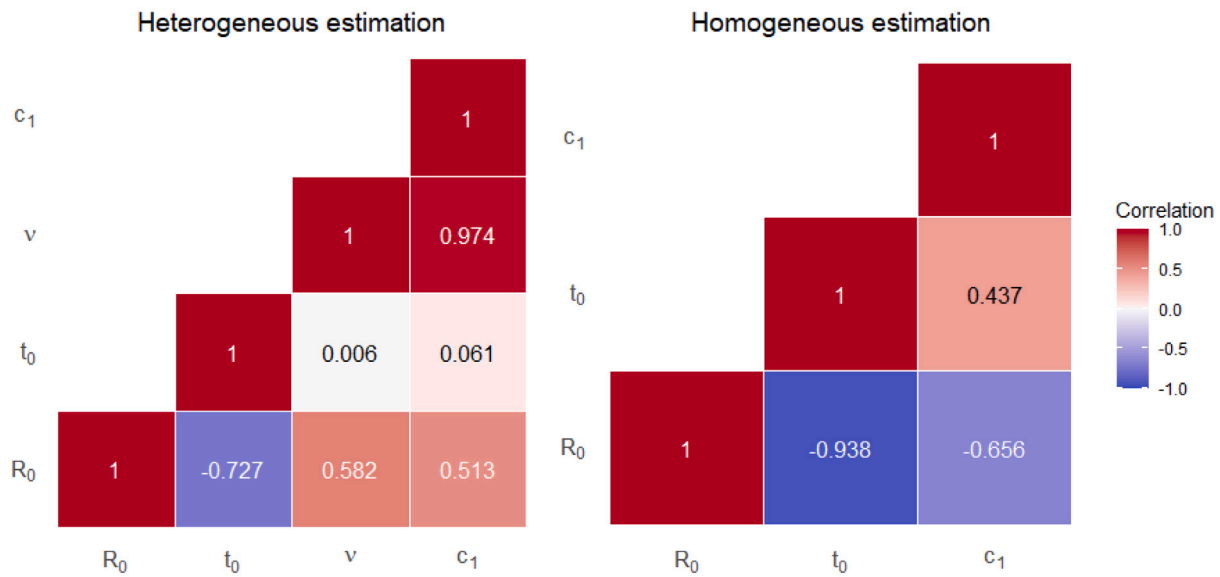


Fig. 8. Median parameter correlation heatmaps for D4 datasets: heterogeneous data with impact of NPIs. Correlations between pairs of estimated parameters are provided in scenarios where fittings are performed with a heterogeneous model (M4, left) or with a homogeneous model (M3, right). M4 results in a very strong positive correlation between ν and c_1 (0.974), whereas in the M3 case there is a strong negative correlation between R_0 and t_0 (−0.938).

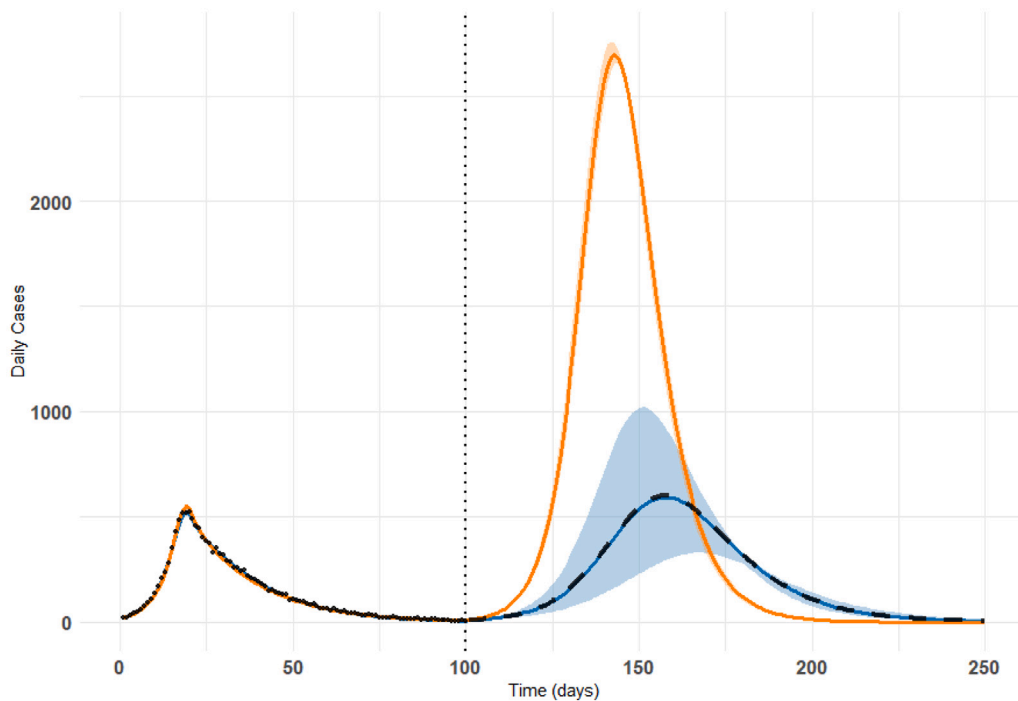


Fig. 9. Prediction trajectories for scenario D4: Heterogeneous data with NPIs. Models were fitted to the first 100 days of data (black dots) and then used to forecast the next 150 days following the release of NPIs at day 101. Heterogeneous model fit (blue); homogeneous model fit (orange). Lines are median trajectories and shaded areas represent 95% CI. The dashed black line represents the “true” trajectory generated with the assumed parameter values. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Here we conduct the first systematic simulation-based investigation of how two-epidemic analyses can resolve identifiability problems inherent to single-epidemic inference as exposed in Section 5. We seek to evaluate: whether analysing simultaneously two epidemics which differ only in epidemic “age” at the start of NPIs improves identifiability; and how achieved improvements depend on the strength of NPIs and the difference in epidemic age when NPIs begin. This is conducted by systematically assessing the degree to which estimation precision and parameter correlations are affected.

In the sections that follow, we present detailed profile likelihood and correlation analyses that demonstrate how this approach breaks the confounding between heterogeneity and intervention effects, among other benefits.

6.1. Eigenvalue and Hessian analysis

For our parameter vector $\theta = (R_0, \nu, t_0, c)$, the Hessian matrix at the maximum likelihood estimate (MLE) represents the local curvature

of the negative log-likelihood function. To examine parameter identifiability, we examine the structure of the Hessian matrix. The eigen decomposition of this Hessian provides information

$$\mathbf{H} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^T, \quad (11)$$

where $\mathbf{\Lambda} = \text{diag}(\lambda_1, \lambda_2, \dots, \lambda_p)$ contains eigenvalues ordered by magnitude ($\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_p > 0$) and $\mathbf{V} = [\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_p]$ contains the corresponding eigenvectors.

The eigenvalues indicate how quickly the likelihood function changes when moving in different directions in parameter space. Large eigenvalues indicate directions with high curvature, where parameters are well-constrained, while small eigenvalues signal directions with low curvature where parameters are poorly constrained. In addition, the condition number $\kappa = \lambda_1/\lambda_p$ indicates numerical stability, with large values ($\kappa > 1000$) suggesting practical identifiability issues.

We performed an analysis for synthetic single epidemics, and then pairs of epidemics which share the same parameters except for the timing of NPIs in terms of epidemic age, with a view to alleviating the correlation as we shall see shortly. In our analysis of the Hessian for single-epidemic fits, we consistently observe one eigenvalue significantly smaller than the others, indicating a direction in parameter space with low curvature and thus poor identifiability with a large condition number. However, if two epidemics curves are simultaneously used to compute the Hessian, the condition number can be substantially reduced.

The correlation matrix derived from the inverse Hessian,

$$\text{Corr}_{ij} = \frac{(\mathbf{H}^{-1})_{ij}}{\sqrt{(\mathbf{H}^{-1})_{ii}(\mathbf{H}^{-1})_{jj}}}, \quad (12)$$

quantifies the statistical dependencies between parameter estimates.

6.2. Profile likelihood analysis

The profile likelihood approach offers a practical method to visualize identifiability issues by exploring how the likelihood changes when one parameter is varied while all others are optimized. For a parameter θ_i , the profile likelihood is defined as

$$PL(\theta_i) = \max_{\theta_j, j \neq i} L(\theta). \quad (13)$$

This approach involves fixing one parameter at various values within a plausible range and re-optimizing all other parameters through maximum likelihood estimation. This process generates a curve of maximum log-likelihood values as a function of the fixed parameter, enabling the construction of confidence intervals and revealing structural correlations between parameters.

We applied this method to 200 simulated datasets using the reduced SEIR model with gamma-distributed susceptibility. Each dataset was generated with parameters $\mathcal{R}_0 = 3.0$ (basic reproduction number), $\nu = 1.414$ (coefficient of variation), $t_0 = 15$ days (behavioural change), and $c_1 = 0.3$ (1– intervention strength), then fitted using maximum likelihood estimation. For the two-epidemic scenario, we simultaneously simulated a “baseline” epidemic (initial conditions $E(0) = 100$, $I(0) = 40$) and an “auxiliary” epidemic ($E(0) = 1000$, $I(0) = 400$) with identical parameters. The different initial conditions account for the difference in epidemic ages at each time point. For each dataset, we profiled all four parameters and analysed the resulting likelihood profiles, parameter correlations, confidence intervals, and eigenvalue structures of the Hessian matrices. During optimization, we employed parameter transformations (logarithmic for $\mathcal{R}_0$, ν , and t_0 ; logit transformation for c_1) to ensure that parameter constraints were maintained during optimization. Confidence intervals were determined using the chi-square cutoff method, where parameter values yielding likelihood ratio statistics below the threshold defined by $\chi^2_{0.95,1} = 3.84$ were included in the 95% confidence interval.

Table 4

Parameter estimates and confidence interval comparisons ($c_1 = 0.3$).

Parameter	Single epidemic		Two epidemics		CI width reduction
	Mean (SD)	CI width	Mean (SD)	CI width	
$\mathcal{R}_0$	3.00 (0.03)	0.090	3.00 (0.01)	0.038	57.6%
ν	1.42 (0.18)	0.571	1.41 (0.01)	0.043	92.5%
t_0	14.98 (0.36)	1.174	15.01 (0.19)	0.667	43.2%
c_1	0.30 (0.02)	0.055	0.30 (0.00)	0.008	86.1%

Figures S12–S15, in Supplementary Material Section S6.1, show representative profile likelihood curves for all four parameters from a randomly selected sample dataset, comparing the single-epidemic versus two-epidemic approaches. Table 4 presents the parameter estimation results across all 200 datasets, comparing the single and two-epidemic approaches. The most substantial improvement is observed for the coefficient of variation parameter (ν), with a CI width reduction of approximately 93%. The 86% reduction in CI width for the intervention strength parameter (c_1) is also noteworthy. These width-reduction improvements can be visualized in Supplementary Figure S16 (Supplementary Material Section S6.2) which displays the 95% confidence intervals of parameter estimates from all the datasets, along with the respective median values.

To further examine the statistical performance of both approaches, we analysed the relative bias and coverage probabilities of the confidence intervals across all 200 datasets, as shown in Supplementary Tables S5 and S6. The results demonstrate that the two-epidemic approach not only provides more precise estimates but also exhibits lower relative bias for most parameters.

Finally, a numerical stability analysis conducted in Supplementary Material Section S6.3 concludes that the two-epidemic approach produces a better-conditioned optimization problem, with lower and less variable condition numbers in comparison with the single-epidemic problem.

6.3. Parameter correlation analysis

We computed pairwise parameter correlations between the 200 estimates to understand how the two-epidemic approach affects parameter identifiability. Fig. 10 shows the correlation matrices for both approaches.

While some parameter pairs show reduced correlations, others maintain similar levels, reflecting the complex trade-offs in parameter identifiability. Most importantly, the two-epidemic approach successfully reduces correlations for the parameter pairs that concern this study the most — those that directly impact our ability to separate heterogeneity effects from intervention effects. This provides the mechanistic foundation for the improved precision and identifiability observed throughout this analysis.

6.4. Sensitivity analysis

Sensitivity analysis examines how changes in model parameters affect the model output. For a parameter θ_j , the sensitivity function $S_{ij}(t)$ measures how the observable data at time t changes with respect to that parameter:

$$S_{ij}(t) = \frac{\partial y_i(t)}{\partial \theta_j} \quad (14)$$

where $y_i(t)$ is the observable (daily incidence) for epidemic i at time t . Intuitively, when two parameters have similar sensitivity patterns, compensation effects are created, such that changes in one parameter can be offset by changes in another to produce similar epidemic curves. However, when sensitivity patterns are different (orthogonal), the parameters become more distinguishable.

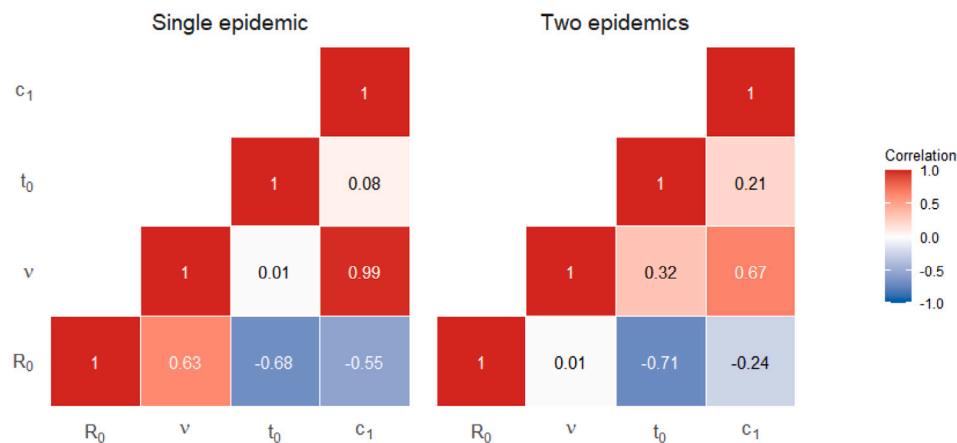


Fig. 10. Median parameter correlation heatmaps comparing single-epidemic (left) and two-epidemic (right) approaches. Lower correlations in the two-epidemic approach indicate improved parameter identifiability.

We conducted a comparative sensitivity analysis for single and two-epidemic approaches. As previously, we simulated epidemic datasets using parameter values $R_0 = 3.0$, $v = 1.414$, $t_0 = 15$ days, $c_1 = 0.3$ (initial conditions, $E(0) = 100$, $I(0) = 40$ for the focal epidemic, and $E(0) = 1000$, $I(0) = 400$ for the auxiliary). We then performed local sensitivity analyses Eq. (14), using finite differences to approximate derivatives of model outputs with respect to parameters (Press et al., 1992).

Figure S18, in Supplementary Material Section S7, illustrates the reduction in parameter compensation as we move from single to two-epidemic approaches. For single epidemics, the sensitivity patterns for CV (v) and intervention strength (c_1) follow nearly identical temporal trajectories, creating identifiability challenges. In contrast, the two-epidemic approach produces orthogonal sensitivity patterns, effectively breaking compensation mechanisms, especially around a time window where natural epidemic dynamics begin to be disrupted by behavioural change and NPIs, since this occurs at different phases for different epidemics. These tests provide further empirical support for the adoption of two-epidemic frameworks as a means to alleviate parameter identifiability issues.

7. Sensitivity to initial conditions of auxiliary epidemic

We now examine how the initial conditions of the auxiliary epidemic affect the gain in parameter identifiability relative to the single-epidemic approach. We present results for three values of the intervention parameter ($c_1 = 0.2, 0.3, 0.4$). All other parameters remain fixed as usual: $R_0 = 3.0$, $v = 1.414$, and $t_0 = 15$ days.

Specifically, we simulated two concurrent epidemics: a focal epidemic with $I(0) = 40$; and an auxiliary epidemic with $I(0) \in \{20, 40, 80, 160, 320, 400\}$. For each scenario, we generated 200 synthetic datasets and analysed parameter correlations using maximum likelihood estimation.

Figs. 11, 12, and 13 show how parameter correlations vary with the initial condition of the auxiliary epidemic.

As expected, all correlations tend to decrease as the initial condition of the auxiliary epidemic deviates from that of the focal (baseline). In addition, we see that as the intervention becomes stronger (decreasing c_1), the correlation between R_0 and v increases, while those between R_0 and t_0 and between v and c_1 decrease. This effect is especially pronounced in the case of v - c_1 .

8. Discussion and conclusions

In this study, we conducted the first systematic investigation of parameter identifiability for the SEIR model with heterogeneity in

susceptibility to infection. The work consisted in simulating datasets using stochastic versions of the model with strategically chosen parameter values – with or without heterogeneity, with or without non-pharmaceutical interventions – and then conducting maximum likelihood estimation to assess how well we could retrieve the assumed parameter values. The key findings are as follows:

- Model parameters are generally estimated accurately, but there is a tendency for high uncertainty in the estimated CV of susceptibility when fitting to data that included NPIs.
- Strong correlations among pairs of estimated parameters are generally apparent, whether heterogeneity is included as a parameter or not, indicating potential issues with identifiability.
- The issues highlighted above can be substantially alleviated by simultaneously fitting two epidemic trajectories with some shared parameters.

The strategy of fitting multiple epidemics with shared parameters to gain identifiability can be extended beyond the idealized scenarios considered here. The reality will be more nuanced. In particular, real data will be noisier, reporting will be delayed and incomplete, exact initial conditions will not be known, distributions of susceptibility may vary between populations, and more parameters will be unknown. The method will need to be adapted creatively for each situation, but the general approach has been used successfully in real-world investigations (e.g., White et al., 2007; Aguas et al., 2008; Gomes et al., 2022). By synthesizing its essence and highlighting its strengths, we hope to motivate further methodological developments and applications of the multiple-epidemic approach featured here.

CRediT authorship contribution statement

Ibrahim Mohammed: Writing – review & editing, Writing – original draft, Formal analysis, Conceptualization. **Chris Robertson:** Writing – review & editing, Writing – original draft, Formal analysis, Conceptualization. **M. Gabriela M. Gomes:** Writing – review & editing, Writing – original draft, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

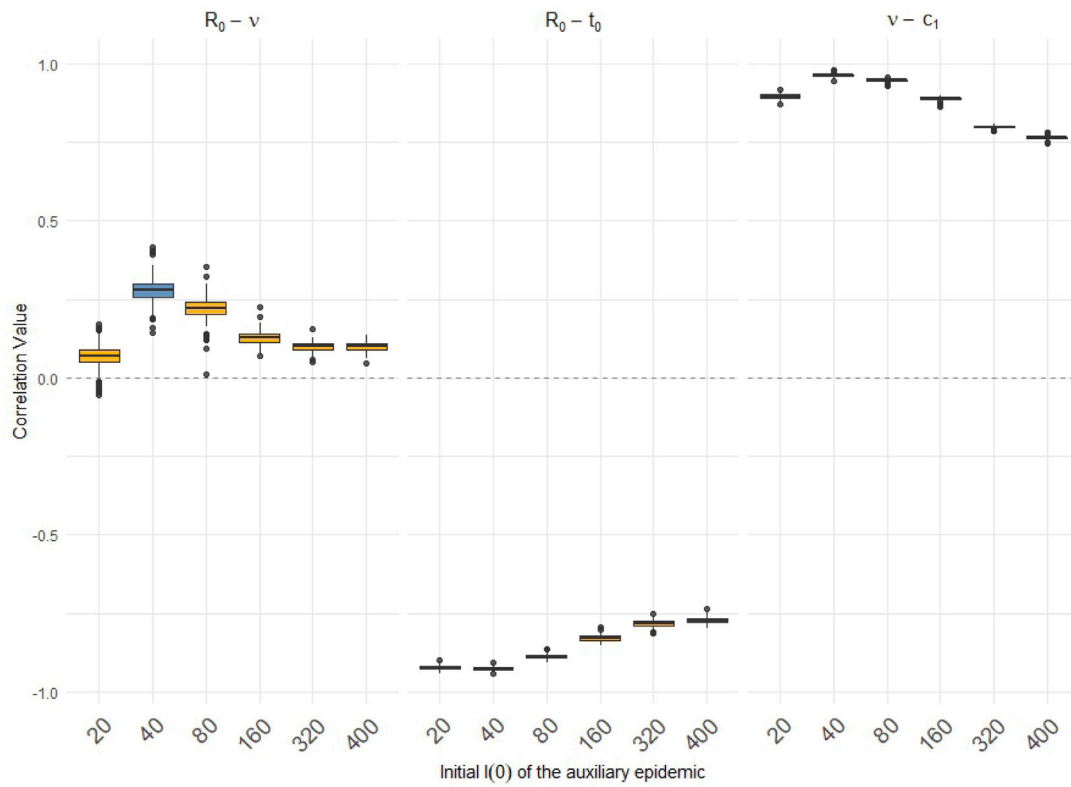


Fig. 11. Correlation distributions in terms of the initial condition of the auxiliary epidemic ($I(0) \in \{20, 40, 80, 160, 320, 400\}$) for intervention parameter $c_1 = 0.4$. The baseline epidemic is initiated with $I(0) = 40$ in all cases. Cases where baseline and auxiliary epidemics coincide (i.e., have the same initial conditions) are marked in blue. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

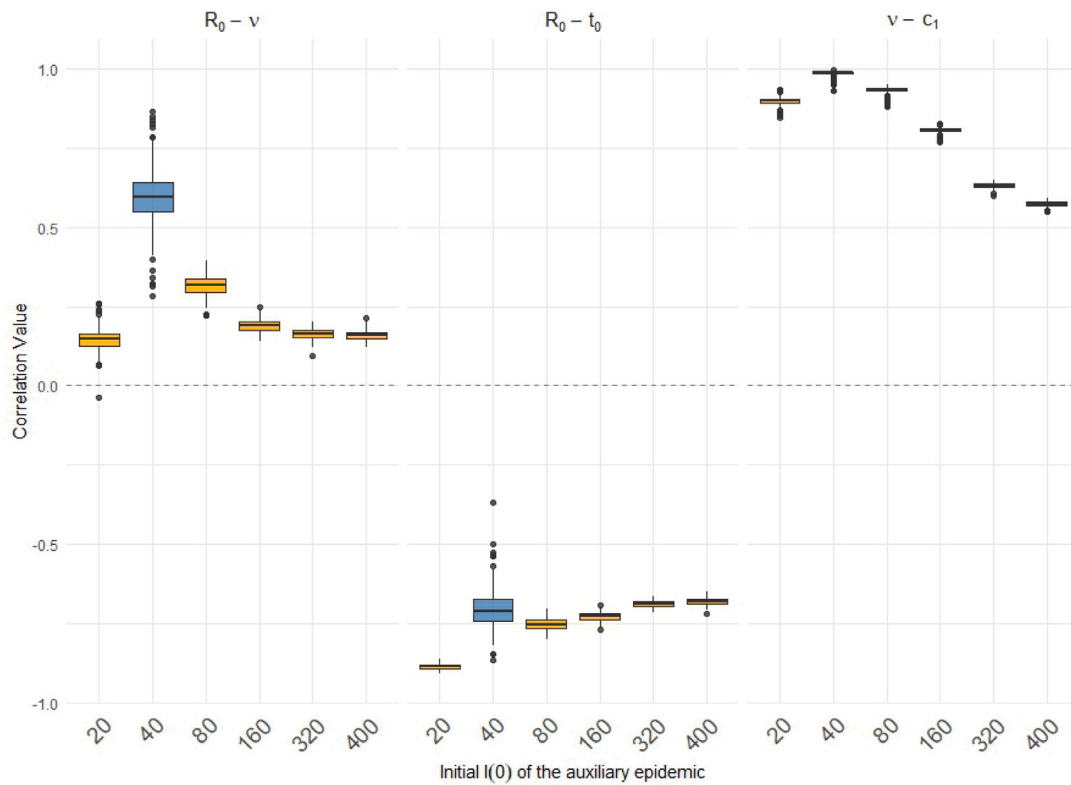


Fig. 12. Correlation distributions in terms of the initial condition of the auxiliary epidemic ($I(0) \in \{20, 40, 80, 160, 320, 400\}$) for intervention parameter $c_1 = 0.3$. The baseline epidemic is initiated with $I(0) = 40$ in all cases. Cases where baseline and auxiliary epidemics coincide (i.e., have the same initial conditions) are marked in blue. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

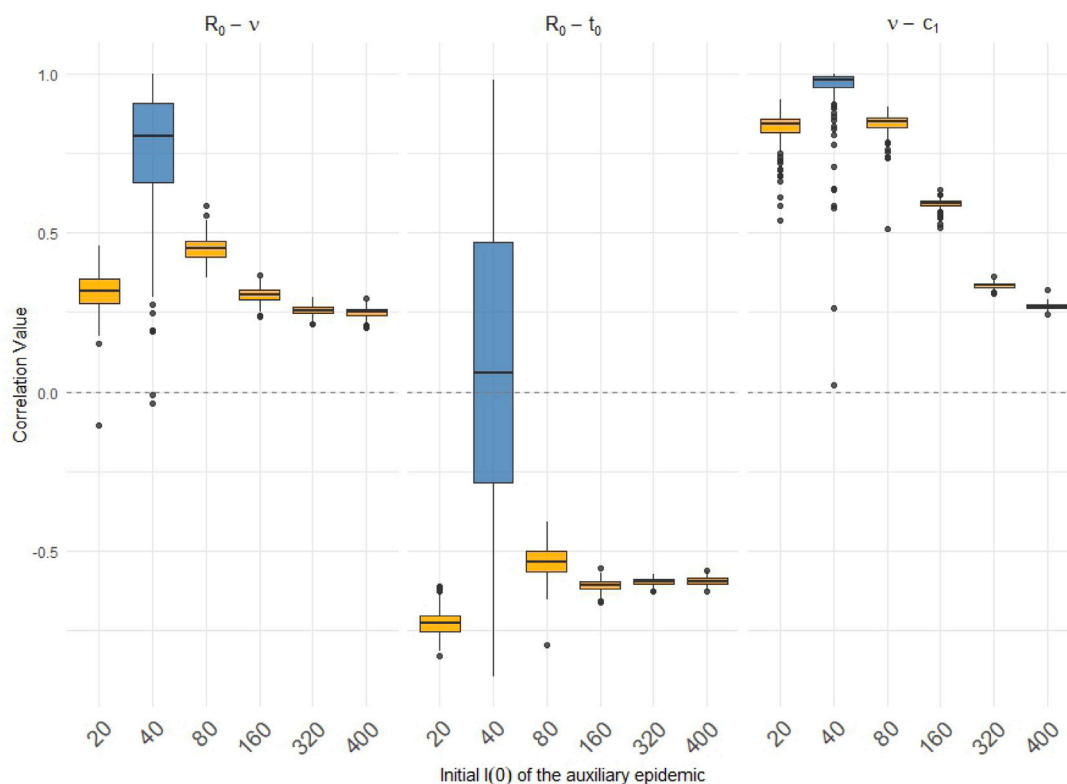


Fig. 13. Correlation distributions in terms of the initial condition of the auxiliary epidemic ($I(0) \in \{20, 40, 80, 160, 320, 400\}$) for intervention parameter $c_1 = 0.2$. The baseline epidemic is initiated with $I(0) = 40$ in all cases. Cases where baseline and auxiliary epidemics coincide (i.e., have the same initial conditions) are marked in blue. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.epidem.2026.100911>.

Data availability

Computer code used to simulate epidemics and estimate model parameters is available from <https://github.com/ibrahimStrathclyde/On-the-simultaneous-inference-of-epidemic-susceptibility-distributions-and-NPI>.

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